Solution

Equation of ellipse is $x^2 + 4y^2 = 32$

$$\frac{x^2}{32} + \frac{y^2}{8} = 1$$

$$a^2 = 32,\ b^2 = 8$$

$$a = 4\sqrt{2},\ b = 2\sqrt{2}$$

Equation of tangent at $\theta = \frac{\pi}{4}$ is

$$\frac{x\cos\frac{\pi}{4}}{4\sqrt{2}} + \frac{y\sin\frac{\pi}{4}}{2\sqrt{2}} = 1$$

$$\frac{x}{8} + \frac{y}{4} = 1$$

$$x + 2y - 8 = 0.$$

Equation of normal is $\frac{4\sqrt{2}x}{\cos\frac{\pi}{4}} - \frac{2\sqrt{2}y}{\sin\frac{\pi}{4}} = 32 - 8$

That is $8x - 4y = 24$

$$2x - y - 6 = 0.$$

Aliter

$$\text{At, } \theta = \frac{\pi}{4},$$

$$(a\cos\theta,\ b\sin\theta) = \left(4\sqrt{2}\cos\frac{\pi}{4},\ 2\sqrt{2}\sin\frac{\pi}{4}\right)$$

$$= (4, 2)$$

$\therefore$ Equation of tangent at $\theta = \frac{\pi}{4}$ is same at $(4, 2)$.

Equation of tangent in cartesian form is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$

$$x + 2y - 8 = 0$$

Slope of tangent is $-\frac{1}{2}$

Slope of normal is 2

Equation of normal is $y - 2 = 2(x - 4)$

$$y - 2x + 6 = 0.$$

EXERCISE 5.4

1. Find the equations of the two tangents that can be drawn from $(5, 2)$ to the ellipse $2x^2 + 7y^2 = 14$.
2. Find the equations of tangents to the hyperbola $\frac{x^2}{16} - \frac{y^2}{64} = 1$ which are parallel to $10x - 3y + 9 = 0$.
3. Show that the line $x - y + 4 = 0$ is a tangent to the ellipse $x^2 + 3y^2 = 12$. Also find the coordinates of the point of contact.

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